

Temperature Kills Models, Not Answers: Decoding Robustness Is a Model Property

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Abstract

Truncation samplers such as top- p , min- p , and top- $n\sigma$ exist to prevent the accuracy collapse of high-temperature sampling. We ask whether the collapse itself is universal. Using a single controlled pipeline, we run seven instruction-tuned models from five families (1.7B to 12B parameters) through a factorial grid of 4 quantization levels, 8 sampler arms, and 2 tasks, plus a temperature-only cell on two further Llama generations, about 204,000 generations in total. At deployment temperatures ($T \leq 1.3$), sampler choice barely matters: outside one family, the per-model accuracy spread across samplers is 1.3 to 10.0 percentage points (pp), and even that overstates the true differences. The exception is the Llama family. Under plain temperature sampling at $T=1.3$, Llama-3.1-8B loses 34 pp (GSM8K) and 41 pp (MMLU-Pro), Llama-3.2-3B collapses the same way, and Llama-3-8B loses 17 pp on MMLU-Pro; every model outside the family stays within 8 pp. The collapse is a termination failure: generations derail midway and never reach an answer, while accuracy conditional on producing a well-formed answer stays at the greedy level in every cell (97% for Llama-3.1-8B on GSM8K at $T=1.3$). A temperature ladder up to $T=2.0$ locates each model’s cliff and shows that truncation samplers only begin to matter past it. A logit probe and a forced-token perturbation experiment trace the fragility to a self-reinforcing instability: one off-distribution token in context multiplies the fragile model’s chance of drawing the next one. Practitioners tuning a current model can leave the sampler alone.

1 Introduction

Most deployed language models generate text by sampling. At each step the model assigns a probability to every candidate next token; the decoder either takes the most probable one (greedy decoding) or draws at random from the distribution. Systems sample because greedy decoding returns one

fixed continuation per prompt, and in open-ended text that continuation tends to be repetitive (Holtzman et al., 2020). A temperature parameter scales the randomness of the draw: raising it moves probability from the model’s preferred tokens toward rare ones.

The rare tokens are where sampling goes wrong. A model’s probability estimates are least reliable in the tail of its distribution, so a high-temperature draw can select a token that fits the context poorly (Holtzman et al., 2020; Hewitt et al., 2022). The selected token then becomes context for every later step, and the continuation can degenerate into incoherent text. Truncation samplers exist to prevent this degeneration: they delete the tail of the distribution before the draw and renormalize what remains. The rules differ in where they cut the tail; top- k , nucleus (top- p), min- p , and top- $n\sigma$ sampling are the four we test, and Section 2 describes each (Fan et al., 2018; Holtzman et al., 2020; Nguyen et al., 2025; Tang et al., 2024). All of these were developed for open-ended generation, such as story writing and dialogue, where output variety is the goal (Garces Arias et al., 2024).

The same samplers are now evaluated on tasks with one checkable answer. Deployed assistants keep sampling on for every request, so math word problems and multiple-choice exams are also decoded at nonzero temperature (Song et al., 2025; Renze and Guven, 2024). On such tasks a decoding method is scored by accuracy: each generation is right or wrong, where open-ended generation needs a human or model judge (Garces Arias et al., 2024). Min- p , for example, reports accuracy gains over top- p on GSM8K at high temperature (Nguyen et al., 2025).

Those accuracy claims drew a critique. Schaeffer et al. (2025) asked whether any one sampler beats the others once statistical controls are in place. Whichever sampler wins that comparison, the comparison itself takes the collapse for

granted: it assumes that raising the temperature eventually ruins every model’s accuracy, and asks only which sampler limits the damage. We test the premise: **is the collapse itself a property of decoding, or of the model?**

Across nine instruction-tuned models from five families, run through one controlled pipeline, the answer is: of the model. The six models from outside the Llama family hold their accuracy at deployment temperatures under every sampler we test, staying within 8 points of their low-temperature scores. Every larger loss belongs to a Llama: Llama-3.1-8B and Llama-3.2-3B lose 34 to 41 accuracy points under plain temperature sampling, and Llama-3-8B loses 17 on the longer task. Sampler choice, the variable the truncation literature tunes, is worth at most a few points anywhere below each model’s collapse temperature.

Our contributions:

- **A single-pipeline factorial study.** Seven instruction-tuned models, four GGUF quantization levels (model weights stored at reduced precision) plus a BF16 anchor, six decoding methods plus two ablation arms that swap the order in which temperature and truncation are applied, and $T \in \{0.7, 1.0, 1.3\}$, on GSM8K and MMLU-Pro ($N=50$ items, $K=3$ repetitions): about 204,000 generations, with item-clustered bootstrap confidence intervals throughout, plus a follow-up temperature-only cell on two more Llama generations.
- **Robustness is model-bound, and the fragility runs in one family.** Every collapse we observe is a Llama, across three of its generations (Section 4.1). On the robust models the sampler is worth at most a few points, and Section 4.2 shows even those few points overstate the true differences.
- **The collapse is a termination failure.** Conditional on emitting a well-formed final answer, accuracy is flat in temperature everywhere. What varies is the probability of derailling into unparseable text mid-generation (Section 4.3).
- **A cliff map from $T=1.3$ to 2.0.** Every model collapses somewhere; what distinguishes models is where. On GSM8K, Llama collapses by $T=1.5$, Qwen2.5 by 1.7, and

Gemma-3 by 2.0, and truncation samplers shift each cliff right in order of tail-cutting strength ($\text{top-}n\sigma \gg \text{min-}p \gg \text{top-}p$). The gains the sampler papers report are real, but they appear past the cliff, above deployment temperatures (Section 4.4).

- **Supporting results.** Quantization from Q8 down to Q3 causes no collapse on any model and does not change which sampler ranks best (Section 4.5). A logit probe and a forced-token experiment (Section 4.6) locate the fragility: every model draws rare off-distribution tokens at the same per-token rate, but one injected tail token multiplies Llama’s subsequent escape hazard 2.3-fold while every robust model absorbs it.

2 Related work

Truncation samplers. Top- k sampling draws from only the k most probable tokens; it was introduced for story generation, where likelihood-maximizing decoding produced generic and repetitive text (Fan et al., 2018). Holtzman et al. (2020) showed that a fixed k fits poorly because the next-token distribution is sometimes peaked and sometimes flat, and proposed nucleus sampling, which keeps the smallest set of tokens whose probabilities sum to a target p . Min- p keeps only the tokens whose probability is at least a set fraction of the top token’s, so the cut is tight where the model is confident and loose where it is uncertain (Nguyen et al., 2025). Top- $n\sigma$ cuts in logit space instead, keeping tokens within n standard deviations of the top logit; temperature rescales all logits equally, so the surviving set does not change with temperature (Tang et al., 2024). Further refinements of the cut exist, among them typical sampling and epsilon and eta sampling (Meister et al., 2023; Hewitt et al., 2022). All of these rules target the same failure: incoherent text drawn from the tail at high temperature. Welleck et al. (2020) proved the worst case of that failure, showing that standard decoders can place probability on generations that never terminate. In contrast to the sampler papers, which compare truncation rules against each other, we test whether the failure they all guard against occurs at deployment temperatures at all. Outside the Llama family it does not, and on the Llamas it takes the non-terminating form: generations run to the token cap without reaching an answer (Section 4.3).

Reproducibility of sampler gains. Schaeffer et al. (2025) reanalyzed min- p on nine models and 31 temperatures, giving every sampler an equal number of hyperparameter settings, and found its reported gains “largely indistinguishable” from the other samplers’. Part of the gains traced to the order in which temperature and truncation are applied. The order matters because the truncation threshold is computed on whichever distribution the rule sees: applied before temperature, the rule cuts the model’s original, sharp distribution; applied after, it cuts one that temperature has already flattened, so the same nominal setting keeps a different set of tokens. We adopt both lessons: we report statistical uncertainty throughout and run the two chain orders as explicit arms (Section 3). That critique pooled its models together, because its question was about the sampler. Our question is about the model, so we keep every model separate and ask whether the collapse happens on each one. Garces Arias et al. (2024) sweeps decoding hyperparameters on open-ended generation with quality and diversity metrics, while our tasks have one checkable answer. Shi et al. (2024) compare decoding methods on closed-ended tasks and report that the best method changes with the task, the model, and the quantization level; the comparison is again among methods, and it leaves our premise question open. Song et al. (2025) documents per-model gaps between greedy decoding and sampling on reasoning benchmarks and treats the gaps as evaluation noise to report; we treat the gap as the object of study and trace it to a termination failure in one model family.

Robustness to decoding randomness. Renze and Guven (2024) sampled temperatures from 0.0 to 1.6 on multiple-choice problem solving and found no statistically significant effect on accuracy between 0.0 and 1.0. Our results replicate that null inside the deployment band. But the null is pooled across models, and pooling hides the outlier: our per-model ladder shows the heterogeneity beginning just past $T=1.0$, where Llama-3.1-8B falls while every other model holds (Section 4.1). The nearest study is Grover (2026), which also asks what makes a model robust to decoding randomness and concludes that instruction tuning is the primary determinant. Our data refine that conclusion: all seven of our models are instruction-tuned and one still collapses, so instruction tuning cannot be the whole answer. Four design differences

let our grid observe what theirs cannot:

- their sweep stops at $T=1.0$, and the Llama collapse happens between 1.0 and 1.3;
- their robust models score 6 to 30% on synthetic tasks, so there is little accuracy left to lose;
- their instruct-versus-base contrast changes quantization at the same time (4-bit versus full precision), while we sweep quantization as its own axis;
- they test no truncation samplers.

Fastowski et al. (2025) connect the entropy of a model’s token distributions to its temperature sensitivity on factual question answering and report that larger models are more stable. Our logit probe supports the entropy link (Section 4.6). Size does not decide the outcome in our grid, because Qwen3-1.7B holds where the larger Llama-3.1-8B collapses.

3 Experimental setup

The study is a factorial grid over four axes: model, quantization level, decoding arm, and temperature, crossed with two tasks. Every claim in the paper is a comparison between cells of this grid, so everything outside the axes is pinned: one llama.cpp commit, each model’s official chat template rendered client-side, the native completion API, and per-record seeds derived deterministically from the cell identity. An accuracy difference between two cells is then attributable to the axes, up to sampling noise that the bootstrap below quantifies.

Decoding. Each cell runs eight arms: a deterministic greedy anchor, plain temperature sampling, and the four truncation rules of Section 2, each at the setting its source paper uses or recommends: top- p 0.95 (Holtzman et al., 2020), top- k 40 (Radford et al., 2019), min- p 0.05 (Nguyen et al., 2025), top- $n\sigma$ 1.0 (Tang et al., 2024). Because the order of temperature and truncation changes which tokens survive (Section 2), we pin one order, temperature first, across the truncation arms, and add two ablation arms that instead apply temperature last (the llama.cpp default) for top- p and min- p . The other two rules need no ablation arm: temperature scaling preserves the token ranking, so top- k keeps the same set under either order, and top- $n\sigma$ keeps the same set at any temperature by construction (Section 2). Temperatures

$T \in \{0.7, 1.0, 1.3\}$ span the deployment band: the defaults that models and engines ship sit between 0.6 and 1.0 (Llama-3.1’s generation config sets 0.6, llama.cpp defaults to 0.8, the OpenAI and Anthropic APIs to 1.0), so 0.7 and 1.0 bracket common practice and 1.3 sits just above it. Section 4.4 extends the axis to $T=2.0$. Every stochastic arm runs $K=3$ repetitions per item; greedy decoding is deterministic, so $K=1$.

Tasks. We evaluate GSM8K, grade-school math word problems (Cobbe et al., 2021) (0-shot chain-of-thought with the prompt of Meta’s Llama evaluation suite (Grattafiori et al., 2024), numeric exact match), and MMLU-Pro, ten-option multiple-choice questions across 14 categories (Wang et al., 2024) (0-shot chain-of-thought, letter exact match), $N=50$ items each, with parsers faithful to lm-eval-harness. Both tasks ask the model to generate a chain of reasoning and land on one exactly checkable answer, and the study needs both properties: exact-match grading keeps effects of a few points free of judge noise, and a derailment can only show up if there is a long generation to derail. The two tasks differ in generation length, with MMLU-Pro’s chains running roughly twice as long as GSM8K’s, a contrast Section 4.4 uses to test whether longer generations fail earlier. Both tasks grade a generation by parsing a final answer out of the generated text, so the parser decides how a generation that never reaches an answer is graded. We therefore log the parse path of every record (strict final-answer sentence, flexible fallback, or failure) and report the parse failure rate as a metric of its own; a failed parse grades as incorrect. Section 4.3 shows that this metric carries the main result. The frozen MMLU-Pro subset is the head of the test split, and the split is sorted by category, so all 50 items come from one category (business); every within-study contrast holds items fixed, and a stratified replication reproduces the headline result (Section 4.1, Appendix C).

Models and quantization. The grid holds seven instruction-tuned models from five model families: Llama-3.1-8B (Grattafiori et al., 2024), Mistral-7B-v0.3 (Jiang et al., 2023), Qwen2.5-7B (Yang et al., 2024), Qwen3-1.7B/4B/8B (Yang et al., 2025) (thinking off; a thinking-on spot-check in Appendix G), and Gemma-3-12B (Gemma Team, 2025). A follow-up temperature-only cell adds two more Llama generations, Llama-3-8B and Llama-3.2-3B (Sec-

tion 4.1). Each model runs at four quantization levels (Q8_0, Q6_K, Q4_K_M, Q3_K_M), all imatrix GGUF conversions from a single provider so the quantization axis is not confounded by the conversion recipe. A BF16 anchor on Qwen3-4B matches Q8 within noise, which lets Q8 stand in for full precision.

Statistical inference. All confidence intervals come from a nonparametric bootstrap ($B=2000$) clustered on task items: we resample the 50 items rather than individual generations, because the $K=3$ repetitions of one item share that item’s difficulty and treating them as independent would understate the uncertainty. Where a comparison shares items across its two sides, which is everywhere in the grid, the bootstrap is paired.

Hardware. Generation ran on RTX 2080 Ti (11 GB) nodes of a university Slurm cluster and one RTX 5070 Ti (16 GB) workstation for the VRAM-heavy cells and the follow-up runs, with per-architecture CUDA builds of the same pinned commit; the study totals about 204,000 graded generations. Full reproducibility details, including commit and file hashes, are in Appendix K.

4 Results

4.1 The collapse is model-bound

We begin with plain temperature sampling, no truncation: whatever accuracy the intact tail loses as temperature rises is the failure the truncation samplers of Section 2 exist to prevent. Figure 1 gives the drop from $T=0.7$ to $T=1.3$, pooled over quantization levels; Table 1 in Appendix A gives the exact values, and Section 4.5 shows quantization produces no collapse on any model.

The models do not split cleanly into robust and fragile. The Qwen3 family is a tight null, Qwen2.5, Gemma, and Mistral drop 2 to 8 pp with at least one task CI excluding zero, and Llama drops 34 to 41 pp. Three controls rule out the obvious confounds. It is not accuracy headroom: Qwen3-1.7B (74% on GSM8K) sits in Llama’s accuracy regime (78%) with the opposite temperature behavior. Model age does not explain it either: Mistral-7B-v0.3 predates Llama-3.1 and is robust. The item subset is not driving it: the stratified MMLU-Pro replication gives Llama +43.3 pp [31.3, 55.3], Qwen2.5 +6.0, and Gemma +7.3.

A single collapsing checkpoint could still be a quirk of that release rather than a family trait, so a follow-up cell repeats the pure-temperature

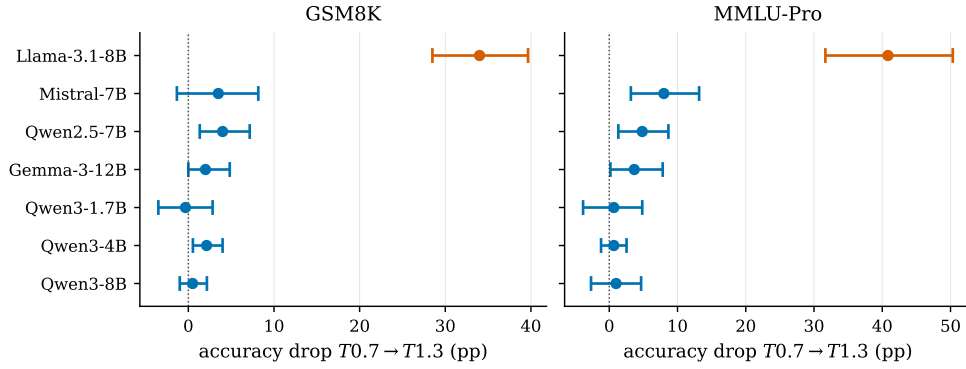


Figure 1: The pure-temperature collapse is model-bound: accuracy drop $T_{0.7} \rightarrow T_{1.3}$, points with 95% item-clustered bootstrap CIs.

axis on two more Llama generations, Llama-3-8B and Llama-3.2-3B (Q8, same frozen items, $K=3$; Table 7, Appendix D). Llama-3.2-3B collapses like Llama-3.1-8B (+38.0 pp [29.3, 47.3] on GSM8K, +36.7 [26.7, 46.0] on MMLU-Pro) and reproduces the degeneration signature of Section 4.3: about half of all generations run to the token cap, and accuracy among well-formed answers stays at or above the greedy level. Llama-3-8B sits between the two: a null on GSM8K (+4.0 [-2.7, 11.3]) but +16.7 pp [9.3, 24.0] on MMLU-Pro, the longer task, a larger drop than any non-Llama model shows in Figure 1. Three Llama generations produce the three largest MMLU-Pro drops in the study, so the fragility is a family trait; what changed in the family’s post-training is the open question of Section 5.

4.2 Sampler choice barely matters before the cliff

The previous subsection varied temperature with the sampler fixed; this one holds temperature at $T=1.3$, the top of the deployment band, and varies the sampler, the axis the truncation literature tunes. The maximum minus minimum accuracy across the seven stochastic sampler arms is 1.3 to 10.0 pp on every robust model (Figure 3 and full grids in Appendix A), and even those numbers overstate the axis: seven equally good samplers would still show a positive spread, because the maximum picks the estimate that got lucky and the minimum the one that did not. The true sampler differences are smaller still. Llama’s spread of 36.7 pp (GSM8K) and 41.8 pp (MMLU-Pro) measures something else. Its arms range from plain temperature, which has already collapsed at $T=1.3$, to top- $n\sigma$, which truncates hard enough to

hold (Section 4.4), so on Llama the spread records how much of the collapse each rule prevents.

Chain order (temperature first versus last) is a no-op on the robust models ($\pm 1-2$ pp). On Llama at $T=1.3$ it does matter: temperature-last truncation is more protective (top- p 64% $\rightarrow$ 73% on GSM8K), as the threshold mechanics of Section 2 predict; both orders are in Appendix F.

4.3 The collapse is degeneration, not wrong answers

An accuracy drop can come from two failures: completed answers that are wrong, or generations that never reach an answer. The parse path that Section 3 logs for every record separates the two, so we decompose Llama’s collapse along it (Figure 4 and full decomposition in Appendix B). Nearly all of the drop is the second failure: at $T=1.3$, 53.8% of Llama’s GSM8K generations and 89.2% of its MMLU-Pro generations derail into token-level noise and run to the token cap. Among the generations that do produce a well-formed final answer (a strict parse: the generation ends with the pinned final-answer sentence), Llama still scores 97.3% on GSM8K at $T=1.3$. That number needs one correction before it can be taken at face value: survival is selective, because easier items produce shorter generations and shorter generations have less room to derail. The fair baseline is therefore greedy decoding restricted to the same surviving items, which scores 95.8% (against 76.0% over all items). Conditional on surviving, then, accuracy at $T=1.3$ matches greedy decoding (97.3 vs. 95.8% on GSM8K; 76.9 vs. 78.8% on MMLU-Pro).

The two tasks disguise one failure mode differently: degenerate text almost always contains a

number, so the GSM8K flexible last-number fallback grades a derailed generation as a wrong answer while the MMLU-Pro letter parser reports a parse failure. The underlying event is the same: the generation derails before it reaches an answer.

4.4 The cliff map: robustness is cliff location

Section 4.1 leaves open whether the robust models lack the failure or merely place it beyond the deployment band, so we extend the temperature axis to $T \in \{1.3, 1.5, 1.7, 2.0\}$ on three models spanning the gradient of Figure 1: Llama-3.1-8B, Qwen2.5-7B, and Gemma-3-12B (Q8, both tasks, $N=50$, $K=3$). Figure 2 plots the accuracy trajectories; Table 8 in Appendix E gives each model’s drop against its own $T=0.7$ anchor. The answer is the second: every model has a cliff, and what distinguishes the models is where it sits. On GSM8K the three models collapse in their Figure 1 order, with cliffs roughly 0.2 apart in temperature, and the confidence intervals separate as the ladder climbs: from $T=1.7$ the three models’ drops are pairwise disjoint. Within each model the cliff sits earlier on MMLU-Pro, the task with roughly twice the generation length, exactly what a compounding per-token derail risk predicts: there Llama has already collapsed at $T=1.3$, and Qwen2.5 and Gemma fall together at $T=1.5$.

Truncation shifts the cliff right in order of tail-cutting strength: top- p buys about 0.2 in temperature, min- p holds 70% at $T=1.5$ on Llama GSM8K where plain temperature is at 9%, and top- $n\sigma$ is flat to $T=2.0$ on all three models, consistent with its construction: the surviving token set is temperature-invariant. The strict-parse rate tracks accuracy almost one-to-one along the whole ladder, so the cliff *is* the degeneration cliff. This reconciles the sampler papers with our null result: min- p and top- $n\sigma$ are validated at temperatures of 1.5 to 3, past every model’s deployment range, where their gains are real and ordered top- $n\sigma \gg$ min- $p \gg$ top- p (the $T=2-3$ probe in Appendix E).

4.5 Quantization and answer agreement

The results so far pool over the four quantization levels. For that to be safe, quantization must not cost accuracy on its own, and it must not change what temperature and the samplers do. This subsection checks both. The direct cost is small. From Q8_0 down to Q3_K_M, greedy accuracy is unchanged on every model of 7B parameters or more: all confidence intervals span zero (Figure 6, Ap-

pendix H). A real cost appears only on the two smallest models and grows as the model shrinks, 7 pp [1, 13] on Qwen3-4B and 9 pp [0, 18] on Qwen3-1.7B.

The interaction is what pooling actually depends on, so we recompute the plain-temperature drop of Section 4.1 separately at each quantization level (Table 11, Appendix H). The result is the same at every level: Llama-3.1 collapses everywhere, with drops of 30 to 47 pp across levels and tasks, while no robust model exceeds 12 pp at any single level. The BF16 anchor on Qwen3-4B, running at full precision, shows the same null drop as its quantized versions, and Llama’s sampler ordering at $T=1.3$ is identical at all four levels. Quantization shifts baseline accuracy on small models; it does not create the collapse, remove it, or change which sampler helps.

A second check addresses a different worry: that the robustness of Section 4.1 is nothing more than determinism. Flat accuracy across temperatures is also what we would see if temperature never really changed the output, because the model’s next-token distribution is peaked enough that sampling returns nearly the greedy path every time. The $K=3$ repetitions of the matrix distinguish the two readings: each item carries three independently sampled generations under identical settings. If the robust models were merely deterministic, the three generations would be near copies of one another. Instead, the token sequences differ across repetitions, so temperature does change what these models write. The question is whether it also changes their answers. For each item we count the distinct parsed answers among its three repetitions, 1 meaning all three agree and 3 meaning all three disagree; this is the agreement that self-consistency decoding (Wang et al., 2023) exploits by majority vote, used here as a diagnostic.

On the robust models the answers converge: 1.1 to 1.4 distinct answers per item on GSM8K at $T=1.3$, and raising temperature from 0.7 to 1.3 moves the statistic by at most +0.17 (Table 12, Appendix I). These models sample genuinely different token sequences and still land on the same answer. Llama-3.1 rises from 1.5 to 2.3 under the same change: its three repetitions increasingly disagree with one another, as expected when each one derails independently. Mistral sits near 2.2 at both temperatures, and its scatter has a different source: at a greedy GSM8K accuracy of 31%,

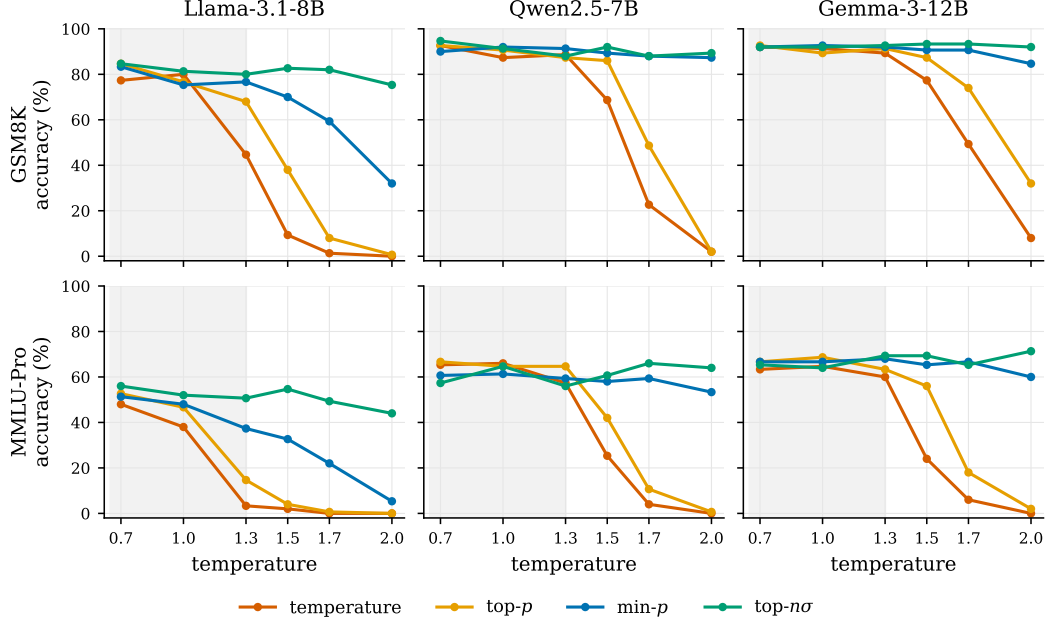


Figure 2: The cliff map. Accuracy versus temperature per sampler (the $T \leq 1.0$ points come from the main matrix; all at Q8). Every model has a cliff; where it sits is the model property. Truncation shifts the cliff right by tail-cutting strength.

most repetitions are wrong, and wrong repetitions rarely agree on the same wrong number at any temperature. The shift attributable to temperature is $+0.27$.

4.6 Toward a mechanism: per-token uncertainty ranks with fragility

If the collapse is a derailment process, fragile models should carry more per-token uncertainty before any derailment happens. To measure that uncertainty we decode the 100 frozen prompts (the 50 items of each task) greedily on every model at Q8 and record the raw top-20 next-token probabilities at each position. Each prompt is summarized by four statistics of these distributions: an entropy lower bound (only the top 20 probabilities are observed), the top-1 probability and its margin over the top-2, and the fraction of positions where the top-1 probability falls below 0.5. We decode greedily so that every model is measured on its own modal path; the statistics then describe the undisturbed trajectory, unaffected by any sampled derailment.

Mean greedy-path entropy ranks the seven models' temperature drops at Spearman $\rho = +0.96$ (flat-position fraction $+0.96$; top-1 margin -0.96 ; Figure 7, Appendix J): the flatter a model's next-token distributions on its own greedy path, the more accuracy it loses to temperature. The sum-

maries cannot separate one pair, though. Llama-3.1 and Mistral are nearly tied on every statistic we record, 0.33 versus 0.32 nats of entropy and 1.0×10^{-3} probability mass beyond the top 20 for both, yet Llama drops 40.8 pp on MMLU-Pro and Mistral 8.0. Uncertainty on the undisturbed path predicts the ordering of the drops and leaves their size unexplained.

Generation lengths rule out one candidate mechanism, derail-and-recover, in which the model wanders off for a stretch, finds its way back, and loses accuracy only when the detour corrupts the answer: a recovered detour leaves extra tokens behind, so terminated generations at $T=1.3$ should be longer than at $T=0.7$. On every robust model their length distribution is instead essentially unchanged (90th-percentile ratio 0.85 to 1.19 across models and tasks; Table 4), and Llama's surviving generations at $T=1.3$ are *shorter* than at $T=0.7$ (90th percentile $\times 0.73$ on GSM8K, $\times 0.55$ on MMLU-Pro). Shortening is the survivorship of Section 4.3 at work: when derailment is permanent and its per-token risk compounds, the generations that still finish are disproportionately the short ones. In every model, a generation that derails does not come back.

That per-token risk can be estimated from the probe itself. From the recorded top-20 arrays we compute, for each position, the probability that

sampling at $T=1.3$ picks a token from beyond the model’s own top 20. We call this the escape hazard: the per-token probability of a genuine tail draw. Llama’s is 0.0033 per position and Mistral’s 0.0034, about 0.85 expected tail draws per 256 positions for both, and across the seven models the hazard ranks the drops at the same $\rho = +0.96$. The two models therefore take their first off-distribution token equally often, and temperature scaling does not reveal a difference the raw probabilities hid. What separates Llama from Mistral must then be what happens after an off-distribution token has entered the model’s own context.

A perturbation experiment measures that directly. For each model and frozen prompt we greedy-decode a 48-token prefix, replace the next token with the model’s own rank-20 candidate at that position (median raw probability 2×10^{-9} , a genuine tail draw), and continue greedily for 256 tokens, alongside a control continuation of the unedited prefix. Judged by the text alone, the forced token is absorbed: the continuations stay locally coherent, and Llama’s reach the token cap no more often than its controls (14.0% for both on GSM8K). The instability appears in the escape hazard over the 32 positions after the injection: Llama’s rises to 2.3 times its baseline (+0.0042 per position), Mistral’s to 1.35 times (+0.0012), and no robust model gains more than +0.0014. This per-model destabilization ranks the temperature drops at $\rho = +0.86$ (the entropy increase, +0.82). It is the compounding loop anticipated in the introduction, measured directly: an off-distribution token in context raises the probability of drawing the next one, and the size of that rise is what distinguishes the fragile models (Appendix J consolidates the per-model probe numbers). With seven models these are consistency checks, not a confirmed causal account, and we report them as such.

5 Discussion

For practitioners. At deployment temperatures the consequential choice is the model. On a robust model every configuration we test stays within a few points of every other, so time spent tuning the sampler buys little; on a fragile model the same sweep spans up to 42 points. Screening for the fragility is cheap, because the collapse is large and needs no judge to detect: fifty items under plain

temperature at the top of the deployment band, and a fragile model reveals itself in the strict-parse rate alone. In production, temperature breaks termination and leaves answer selection intact, so a derailed generation carries no well-formed final answer and simple output validation catches it, while the rate of silent errors, well-formed but wrong answers, barely moves.

For the sampler literature. Truncation samplers solve a real problem, and the problem sits above the temperatures at which models are deployed. The papers introducing min- p and top- $n\sigma$ validate them at $T \geq 1.5$, past every cliff we map, so their reported gains are real safety margin beyond the cliff; our grid shows that below the cliff there is no failure left for a sampler to prevent. A sampler evaluation should therefore say where its models’ cliffs sit, since a gain measured past one model’s cliff need not exist on a model whose cliff lies further out.

For the robustness line. The fragility follows a lineage: three Llama generations produce the three largest MMLU-Pro drops in the study while models older, newer, smaller, and larger stay robust, which points at something the family’s post-training recipe shares. Our contribution to that question is the behavioral signature a mechanism study should explain: termination failure with intact answer selection, first off-distribution tokens that arrive at the same per-token rate in every model, and a post-injection destabilization that only the fragile family shows. Which post-training choices set the size of that destabilization is, to us, the most interesting open question the data raise.

6 Conclusion

Across nine models from five families, the choice of truncation sampler is a second-order effect at deployment temperatures: robust models hold within a few points under every sampler we test, while the Llama family loses up to 41 points under plain temperature across three of its generations. Every model has a cliff above its deployment range, robustness is where the cliff sits, and truncation samplers move the cliff without mattering below it. These results argue for screening temperature fragility before deployment rather than tuning the sampler after it. The open question our data sharpen but cannot answer is what, in post-training, moves the cliff.

Limitations

Our sample is 50 items per task with $K=3$ repetitions per cell; the small item count is what makes a factorial of about 204,000 generations feasible. Item-clustered bootstrap CIs accompany every estimate, and the headline effects are 5 to 10 times their CI widths. Mistral-7B sits near floor on GSM8K (32.7% at $T=0.7$), which limits collapse-detection power in that cell; its MMLU-Pro drop is measured away from floor and its CI excludes zero. The frozen MMLU-Pro subset is single-category (business) because the test split is sorted by category; the stratified $N=50$ replication of Appendix C reproduces the headline result.

The scope is open-weight instruction-tuned models of 1.7B to 12B parameters, served through one inference engine. Whether larger or proprietary models share the Llama family’s fragility is untested. Both tasks are English, zero-shot, and objective: a single exactly checkable answer is what lets the parser measure degeneration. The original use case of truncation samplers, open-ended generation with no checkable answer, lies outside this design, so our null result on sampler gains does not extend to it. Quantization covers llama.cpp k-quants only, one recipe by design so that the quantization axis varies precision and nothing else; other recipes, and other engines’ sampler implementations, may behave differently. Thinking mode is off throughout the main grid; the thinking-on spot-check of Appendix G agrees with it.

Every fragile model we observe is a Llama. The lineage cell of Section 4.1 shows the fragility spans three generations, so it is a family trait rather than a quirk of one checkpoint, but with a single fragile family we cannot say how common such families are.

Three analysis caveats. The max-minus-min sampler spread is biased upward, so it overstates the differences between samplers; the bias is conservative for our claim that those differences are small. Accuracy conditional on a strict parse carries item survivorship, which Section 4.3 quantifies and normalizes. The mechanism probes correlate seven models, and Section 4.6 presents them as consistency checks rather than a causal account.

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A Full sampler-temperature grids

Table 1 gives the exact pure-temperature drops plotted in Figure 1. Figure 3 plots the per-model sampler max-min spread of Section 4.2. Tables 2 and 3 give the full main-matrix grids: accuracy for every sampler arm and temperature, per model, pooled over quantization levels.

Table 1: Pure-temperature accuracy drop $T_{0.7} \rightarrow T_{1.3}$ (pp, 95% item-clustered bootstrap CIs), pooled over quantization levels.

Model	GSM8K	MMLU-Pro
Llama-3.1-8B	+34.0 [28.5, 39.7]	+40.8 [31.8, 50.3]
Mistral-7B-v0.3	+3.5 [-1.2, 8.5]	+8.0 [3.5, 13.2]
Qwen2.5-7B	+4.0 [1.2, 7.3]	+4.8 [1.3, 8.8]
Gemma-3-12B	+2.0 [-0.0, 4.8]	+3.7 [0.2, 7.7]
Qwen3-8B	+0.5 [-1.5, 2.6]	+1.0 [-2.4, 4.8]
Qwen3-4B	+2.1 [0.4, 3.9]	+0.7 [-1.0, 2.5]
Qwen3-1.7B	-0.3 [-3.8, 3.0]	+0.7 [-3.4, 5.0]

B Degeneration decomposition

Figure 4 plots overall accuracy against accuracy conditional on a well-formed answer. Table 5 decomposes accuracy under plain temperature sampling by parse path: the strict final-answer-sentence rate, the parse-failure rate, the fraction of generations that ran to the token cap (including server-recovered degenerate outputs), and accuracy conditional on a strict parse. Figure 5 shows two representative degenerate outputs. The pattern is consistent: a coherent opening, a mid-generation derailment, then token noise until the cap. The model never reaches an answer at all.

Table 4 gives the length-censoring evidence used in Section 4.6: run-to-cap rates and the 90th percentile of completion length among *terminated* generations at $T=0.7$ versus $T=1.3$. Robust models are unchanged (ratio 0.85 to 1.19); Llama’s surviving generations shorten ($\times 0.73$ GSM8K, $\times 0.55$ MMLU-Pro), so derailment is absorbing and survival is decided by whether the generation finishes before the first bad draw.

C Stratified MMLU-Pro replication

The frozen $N=50$ MMLU-Pro head subset is single-category (business) because the test split is sorted by category. Table 6 repeats the pure-temperature axis (Q8, $K=3$) on a deterministic stratified subset with 3–4 items from each of the 14 categories. The collapse ordering reproduces.

D Llama lineage cell

The lineage cell repeats the pure-temperature axis (greedy anchor plus plain temperature at $T \in \{0.7, 1.0, 1.3\}$) on Llama-3-8B and Llama-3.2-3B: Q8_0, both frozen $N=50$ tasks, $K=3$, 2,000 generations, same pipeline, prompts, and seed derivation as the main matrix. Table 7 reports both models next to the Q8-only Llama-3.1-8B slice of the

Table 2: GSM8K accuracy (%) by sampler and temperature, pooled over quantization.

Model	Sampler	$T=0.7$	$T=1.0$	$T=1.3$
Llama-3.1-8B	greedy ($T=0$)	78.0	–	–
	temperature	78.2	74.8	44.2
	top- k	79.8	75.2	58.7
	top- p	81.5	78.0	63.8
	min- p	81.2	78.0	72.0
	top- $n\sigma$	78.8	80.7	80.8
	top- p (temp-last)	80.8	78.7	73.0
	min- p (temp-last)	78.7	81.3	76.2
Mistral-7B	greedy ($T=0$)	31.0	–	–
	temperature	32.7	30.0	29.2
	top- k	31.8	31.7	29.8
	top- p	28.2	31.0	34.2
	min- p	32.3	28.3	28.8
	top- $n\sigma$	29.5	30.7	30.2
	top- p (temp-last)	29.5	31.2	29.5
	min- p (temp-last)	29.5	32.0	30.5
Qwen2.5-7B	greedy ($T=0$)	93.5	–	–
	temperature	91.5	91.7	87.5
	top- k	90.2	88.5	88.7
	top- p	92.2	91.2	90.3
	min- p	90.8	90.5	91.0
	top- $n\sigma$	92.7	91.2	90.7
	top- p (temp-last)	90.8	91.3	90.3
	min- p (temp-last)	92.0	90.7	91.3
Gemma-3-12B	greedy ($T=0$)	92.0	–	–
	temperature	91.0	90.2	89.0
	top- k	91.8	91.7	90.7
	top- p	91.5	90.3	90.7
	min- p	91.5	91.3	92.3
	top- $n\sigma$	91.3	91.5	91.0
	top- p (temp-last)	90.2	91.8	91.5
	min- p (temp-last)	91.2	90.8	90.3
Qwen3-1.7B	greedy ($T=0$)	75.5	–	–
	temperature	73.3	73.7	73.7
	top- k	72.8	72.7	73.3
	top- p	73.8	75.0	72.7
	min- p	72.2	71.7	74.0
	top- $n\sigma$	74.3	73.5	72.3
	top- p (temp-last)	73.0	72.0	71.3
	min- p (temp-last)	74.7	73.3	74.5
Qwen3-4B	greedy ($T=0$)	95.2	–	–
	temperature	95.2	94.0	93.1
	top- k	94.0	94.5	94.1
	top- p	95.5	95.5	93.5
	min- p	95.1	94.7	92.9
	top- $n\sigma$	94.8	93.9	93.9
	top- p (temp-last)	94.8	95.2	94.3
	min- p (temp-last)	94.5	93.9	93.9
Qwen3-8B	greedy ($T=0$)	94.5	–	–
	temperature	95.0	94.2	94.5
	top- k	94.8	94.8	93.5
	top- p	94.0	94.3	93.7
	min- p	94.8	95.3	94.7
	top- $n\sigma$	96.3	94.2	94.3
	top- p (temp-last)	93.8	94.0	94.0
	min- p (temp-last)	96.2	94.7	95.3

Table 3: MMLU-Pro accuracy (%) by sampler and temperature, pooled over quantization.

Model	Sampler	$T=0.7$	$T=1.0$	$T=1.3$
Llama-3.1-8B	greedy ($T=0$)	54.0	–	–
	temperature	47.8	39.7	7.0
	top- k	51.2	43.3	26.8
	top- p	51.3	46.0	16.0
	min- p	51.2	50.8	39.5
	top- $n\sigma$	51.0	50.5	48.8
	top- p (temp-last)	50.3	46.7	37.5
	min- p (temp-last)	51.2	50.2	43.5
Mistral-7B	greedy ($T=0$)	35.5	–	–
	temperature	30.5	30.2	22.5
	top- k	32.5	32.3	28.3
	top- p	30.2	29.5	25.3
	min- p	33.8	30.7	31.0
	top- $n\sigma$	32.3	31.5	31.5
	top- p (temp-last)	29.2	30.2	32.5
	min- p (temp-last)	33.2	33.3	28.7
Qwen2.5-7B	greedy ($T=0$)	64.0	–	–
	temperature	62.5	61.3	57.7
	top- k	62.5	61.7	57.0
	top- p	65.5	62.8	62.7
	min- p	61.3	63.5	59.7
	top- $n\sigma$	62.0	63.5	61.0
	top- p (temp-last)	63.7	62.7	61.3
	min- p (temp-last)	64.2	63.0	62.3
Gemma-3-12B	greedy ($T=0$)	61.0	–	–
	temperature	63.7	66.7	60.0
	top- k	64.3	62.8	63.7
	top- p	65.7	66.7	63.8
	min- p	65.3	67.0	65.5
	top- $n\sigma$	65.2	65.8	65.8
	top- p (temp-last)	65.8	65.0	65.2
	min- p (temp-last)	65.3	63.7	63.8
Qwen3-1.7B	greedy ($T=0$)	45.5	–	–
	temperature	46.0	45.7	45.3
	top- k	47.5	46.5	45.0
	top- p	47.2	47.0	47.2
	min- p	47.0	47.7	46.2
	top- $n\sigma$	46.8	47.2	46.2
	top- p (temp-last)	47.7	46.8	43.5
	min- p (temp-last)	45.7	46.2	49.0
Qwen3-4B	greedy ($T=0$)	61.6	–	–
	temperature	61.9	62.5	61.2
	top- k	63.5	61.1	61.2
	top- p	62.8	61.3	62.8
	min- p	60.9	62.7	62.9
	top- $n\sigma$	62.4	62.4	61.5
	top- p (temp-last)	62.9	62.7	61.7
	min- p (temp-last)	61.3	63.6	61.5
Qwen3-8B	greedy ($T=0$)	60.5	–	–
	temperature	60.5	59.0	59.5
	top- k	60.2	61.3	57.0
	top- p	60.5	61.3	60.5
	min- p	61.0	59.2	59.7
	top- $n\sigma$	60.0	58.8	58.7
	top- p (temp-last)	60.7	59.5	60.5
	min- p (temp-last)	61.3	60.7	59.3

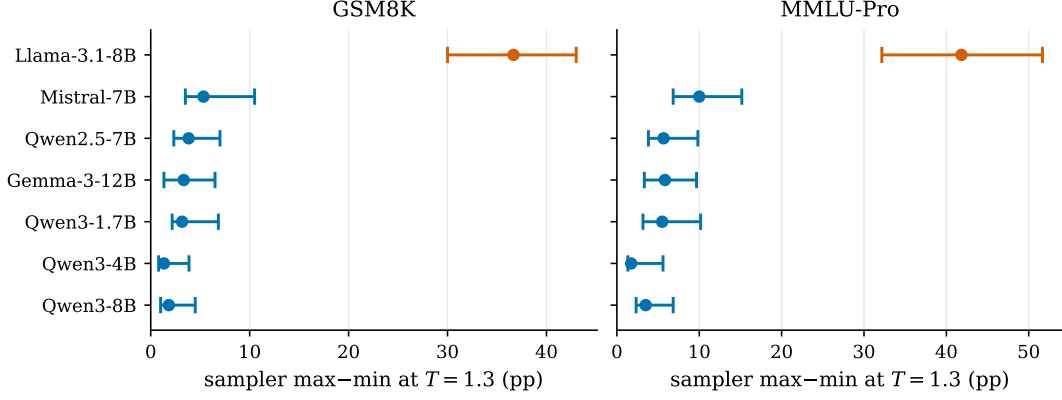


Figure 3: Sampler max-min accuracy spread at $T=1.3$ per model (95% CIs).

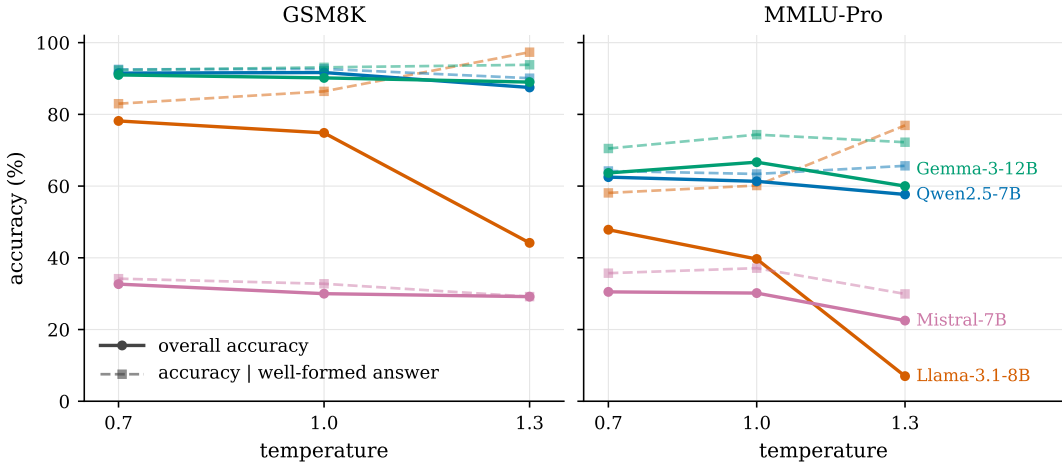


Figure 4: Overall accuracy (solid) versus accuracy among well-formed, strict-parsed answers (dashed) under plain temperature sampling; four models spanning the robustness gradient (the three Qwen3 models are flat and omitted for legibility). The dashed lines stay flat: among well-formed answers, accuracy barely moves with temperature.

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Figure 5: Two representative degenerate outputs (Llama-3.1-8B, Q8, plain temperature, $T=1.3$, MMLU-Pro; both ran to the 1024-token cap).

main matrix, which is why the Llama-3.1 numbers differ slightly from the quantization-pooled Table 1. Greedy anchors are ordinary (Llama-3-8B 76.0% GSM8K / 40.0% MMLU-Pro; Llama-3.2-3B 74.0 / 42.0%). Llama-3.2-3B reproduces the full degeneration signature at $T=1.3$: strict-parse rate $98.7 \rightarrow 41.3\%$ on GSM8K and $82.7 \rightarrow 6.0\%$ on MMLU-Pro, 50 to 51% of generations at the token cap, and accuracy conditional on a strict parse at or above the greedy level (85.5 and 55.6%). Llama-3-8B's MMLU-Pro drop has a milder surface: strict-parse $97.3 \rightarrow 65.3\%$ with only 4.7% at the cap, so its lost generations mostly terminate as parse failures short of the cap.

E High-temperature regime

Table 8 gives each ladder model's plain-temperature drop against its own $T=0.7$ anchor, the values behind Figure 2.

Table 4: Terminated-length censoring under plain temperature at $T=0.7$ versus $T=1.3$ (pooled over quantization). “At cap” includes server-recovered degenerate streams.

Model	Task	at cap (%)		p90 length		ratio
		0.7	1.3	0.7	1.3	
Llama-3.1-8B	GSM8K	5.2	53.8	318	232	$0.73\times$
	MMLU-Pro	11.0	89.2	633	345	$0.55\times$
Mistral-7B	GSM8K	3.5	8.3	367	378	$1.03\times$
	MMLU-Pro	1.0	12.5	628	750	$1.19\times$
Qwen2.5-7B	GSM8K	0.2	1.7	341	361	$1.06\times$
	MMLU-Pro	2.5	10.2	771	732	$0.95\times$
Gemma-3-12B	GSM8K	2.0	5.3	363	344	$0.95\times$
	MMLU-Pro	9.8	17.2	753	763	$1.01\times$
Qwen3-1.7B	GSM8K	2.0	2.2	324	339	$1.05\times$
	MMLU-Pro	15.8	16.3	779	806	$1.03\times$
Qwen3-4B	GSM8K	0.9	0.8	353	361	$1.02\times$
	MMLU-Pro	11.9	12.4	778	731	$0.94\times$
Qwen3-8B	GSM8K	2.3	2.3	379	389	$1.03\times$
	MMLU-Pro	12.3	16.0	850	724	$0.85\times$

Table 9 reports a small probe ($N=15$, $K=3$, Q8) in the regime where $\min-p$ and $\text{top-}n\sigma$ report their largest gains. The sampler ordering is $\text{top-}n\sigma \gg \min-p \gg \text{top-}p \approx \text{temperature}$; the zero cells are entirely degeneration-to-cap, and even the most robust model collapses under plain temperature.

F Temperature-first versus temperature-last

Table 10 gives both chain orders for the two probability-space truncators. The two orders are indistinguishable on the robust models. On Llama at $T=1.3$, truncating before temperature scaling is more protective, worth $+9.2$ pp on GSM8K and $+21.5$ pp on MMLU-Pro for $\text{top-}p$ ($+4.2$ and $+4.0$ for $\min-p$); the results section explains why.

G Thinking-mode spot-check

The matrix runs Qwen3 with thinking off. A confirmatory spot-check (Qwen3-8B, Q8, thinking on, 450 generations, $N=25$, $K=2$) shows the same temperature robustness: under plain temperature, GSM8K 94% at $T=0.7$ versus 96% at $T=1.3$, and MMLU-Pro 56% versus 46% (small sample), with $\min-p$ flat on both tasks. The robustness is not an artifact of suppressing the reasoning mode.

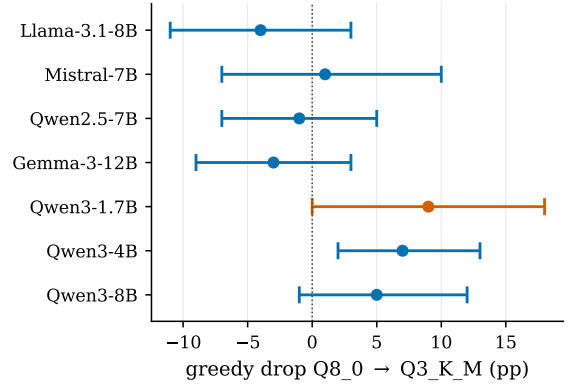


Figure 6: Greedy accuracy drop Q8_0 → Q3_K_M per model (95% CIs), pooled tasks.

H Per-quantization-level temperature drops

Figure 6 plots the direct cost of quantization on greedy accuracy. Table 11 recomputes the plain-temperature drop of Table 1 separately at each quantization level. Each cell pools 150 generations ($N=50$, $K=3$), so single-level estimates are noisier than the pooled ones. The BF16 anchor on Qwen3-4B shows the same null drop as its quantized versions (-1.3 pp GSM8K, -2.0 pp MMLU-Pro).

I Answer agreement across repetitions

Mean distinct parsed answers per item across the $K=3$ repetitions under plain temperature (Table 12). Robust models sample diverse token sequences that converge on the same answer; a parse failure counts as one distinct answer, so the statistic is a divergence proxy. The parse-failure convention censors Llama-3.1 on MMLU-Pro at $T=1.3$: with 84.5% of generations failing to parse, the repetitions agree on having no answer, and the statistic falls to 1.28.

J Mechanism probes

Figure 7 plots greedy-path entropy against the temperature drop. Table 13 consolidates the per-model mechanism evidence of Section 4.6: greedy-path entropy lower bound and flat-position fraction ($\text{top-1} < 0.5$) from the summary probe; the temperature-scaled beyond-top-20 escape hazard per position at $T=1.3$ from the raw top-20 arrays; and the perturbation deltas (post-injection minus control over 32 positions, escape hazard and entropy) from the forced-token experiment. Es-

Table 5: Decomposition of the temperature effect (plain temperature, pooled over quantization; %).

Model	Task	T	acc	strict	failed	at cap	acc strict
Llama-3.1-8B	GSM8K	0.7	78.2	93.0	0.0	5.2	83.0
	GSM8K	1.0	74.8	84.7	0.0	7.8	86.4
	GSM8K	1.3	44.2	43.8	0.5	53.8	97.3
	MMLU-Pro	0.7	47.8	82.3	15.3	11.0	58.1
	MMLU-Pro	1.0	39.7	64.8	29.0	23.7	60.2
	MMLU-Pro	1.3	7.0	8.7	84.5	89.2	76.9
Mistral-7B	GSM8K	0.7	32.7	72.2	0.3	3.5	34.2
	GSM8K	1.0	30.0	67.7	0.0	4.3	32.8
	GSM8K	1.3	29.2	62.8	0.0	8.3	29.2
	MMLU-Pro	0.7	30.5	84.0	12.8	1.0	35.7
	MMLU-Pro	1.0	30.2	80.8	17.0	1.0	37.1
	MMLU-Pro	1.3	22.5	73.5	21.2	12.5	29.9
Qwen2.5-7B	GSM8K	0.7	91.5	98.3	0.0	0.2	92.5
	GSM8K	1.0	91.7	95.7	0.0	0.5	92.7
	GSM8K	1.3	87.5	92.5	0.0	1.7	90.1
	MMLU-Pro	0.7	62.5	97.3	2.3	2.5	64.2
	MMLU-Pro	1.0	61.3	96.5	2.5	3.0	63.4
	MMLU-Pro	1.3	57.7	87.3	10.5	10.2	65.6
Gemma-3-12B	GSM8K	0.7	91.0	98.0	0.0	2.0	92.3
	GSM8K	1.0	90.2	96.8	0.0	3.3	93.1
	GSM8K	1.3	89.0	94.5	0.0	5.3	93.8
	MMLU-Pro	0.7	63.7	90.3	9.0	9.8	70.5
	MMLU-Pro	1.0	66.7	89.7	8.2	10.3	74.3
	MMLU-Pro	1.3	60.0	82.8	14.5	17.2	72.2
Qwen3-1.7B	GSM8K	0.7	73.3	20.3	0.0	2.0	73.8
	GSM8K	1.0	73.7	20.0	0.0	0.7	71.7
	GSM8K	1.3	73.7	19.0	0.0	2.2	71.1
	MMLU-Pro	0.7	46.0	83.7	13.3	15.8	54.8
	MMLU-Pro	1.0	45.7	82.2	15.5	16.3	55.4
	MMLU-Pro	1.3	45.3	83.2	13.5	16.3	54.5
Qwen3-4B	GSM8K	0.7	95.2	94.3	0.0	0.9	95.3
	GSM8K	1.0	94.0	92.3	0.0	0.8	94.8
	GSM8K	1.3	93.1	95.6	0.0	0.8	93.7
	MMLU-Pro	0.7	61.9	87.2	12.5	11.9	70.9
	MMLU-Pro	1.0	62.5	87.3	12.0	10.8	71.6
	MMLU-Pro	1.3	61.2	87.5	11.1	12.4	70.0
Qwen3-8B	GSM8K	0.7	95.0	48.2	0.0	2.3	98.3
	GSM8K	1.0	94.2	49.2	0.0	2.3	98.6
	GSM8K	1.3	94.5	52.2	0.0	2.3	98.1
	MMLU-Pro	0.7	60.5	82.5	15.2	12.3	72.3
	MMLU-Pro	1.0	59.0	80.5	15.5	14.8	72.3
	MMLU-Pro	1.3	59.5	79.8	16.3	16.0	73.5

Table 6: Stratified versus head-of-split (business) MMLU-Pro subset; plain temperature accuracy (%) at Q8_0.

Model	Subset	T			
		0.7	1.0	1.3	drop
Llama-3.1-8B	stratified	44.7	38.7	1.3	+43.3 pp
Llama-3.1-8B	head-50	48.0	38.0	10.0	+38.0 pp
Qwen2.5-7B	stratified	54.7	53.3	48.7	+6.0 pp
Qwen2.5-7B	head-50	65.3	66.0	58.0	+7.3 pp
Gemma-3-12B	stratified	64.0	62.7	56.7	+7.3 pp
Gemma-3-12B	head-50	63.3	64.7	60.0	+3.3 pp

cape columns are $\times 10^{-3}$. Every static column ties Llama-3.1 with Mistral; only the perturbation response separates them.

K Reproducibility

- **Inference stack.** llama.cpp at commit 5aba5364 (server build 9456), native /completion endpoint, CUDA builds of the same commit per GPU architecture (sm_75 for RTX 2080 Ti, sm_120 for RTX 5070 Ti). Server-side BOS insertion is disabled; the client-rendered chat template owns the special tokens, and each record stores the

Table 7: Llama lineage: plain-temperature accuracy and $T_{0.7} \rightarrow 1.3$ drop (pp, 95% item-clustered bootstrap CIs), all at Q8_0. The Llama-3.1-8B rows are the Q8-only slice of the main matrix.

Model	Task	acc (%) at T		drop [95% CI]
		0.7	1.3	
Llama-3-8B	GSM8K	75.3	71.3	+4.0 [−2.7, 11.3]
	MMLU-Pro	44.0	27.3	+16.7 [9.3, 24.0]
Llama-3.1-8B	GSM8K	77.3	47.3	+30.0 [21.3, 38.7]
	MMLU-Pro	48.0	10.0	+38.0 [28.0, 48.7]
Llama-3.2-3B	GSM8K	74.0	36.0	+38.0 [29.3, 47.3]
	MMLU-Pro	40.0	3.3	+36.7 [26.7, 46.0]

Table 8: The cliff ladder: accuracy drop against each model’s own $T=0.7$ anchor under plain temperature (pp, 95% item-clustered bootstrap CIs; Q8).

T	Llama-3.1-8B	Qwen2.5-7B	Gemma-3-12B
<i>GSM8K</i>			
1.3	+32.7 [24.7, 41.3]	+4.0 [−0.7, 9.3]	+2.7 [−2.0, 8.0]
1.5	+68.0 [58.0, 76.7]	+24.0 [15.3, 33.3]	+14.7 [8.7, 21.3]
1.7	+76.0 [66.7, 84.7]	+70.0 [62.0, 77.3]	+42.7 [34.7, 50.7]
2.0	+77.3 [67.3, 86.0]	+90.7 [82.7, 97.3]	+84.0 [76.7, 90.7]
<i>MMLU-Pro</i>			
1.3	+44.7 [34.0, 56.0]	+8.0 [0.0, 16.7]	+3.3 [−3.3, 10.0]
1.5	+46.0 [35.3, 57.3]	+40.0 [28.7, 52.0]	+39.3 [29.3, 48.7]
1.7	+48.0 [36.7, 60.0]	+61.3 [48.7, 74.0]	+57.3 [45.3, 68.0]
2.0	+48.0 [36.7, 59.3]	+65.3 [53.3, 78.0]	+63.3 [51.3, 75.3]

Table 9: Accuracy (%) at $T=2$ and $T=3$ (45 observations per cell).

Task	Sampler	Llama-3.1-8B		Gemma-3-12B	
		$T=2$	$T=3$	$T=2$	$T=3$
GSM8K	temperature	0	0	16	0
	top- p	0	0	27	0
	min- p	33	0	78	24
	top- $n\sigma$	71	80	87	89
MMLU-Pro	temperature	0	0	0	0
	top- p	0	0	0	0
	min- p	0	0	58	0
	top- $n\sigma$	47	51	69	62

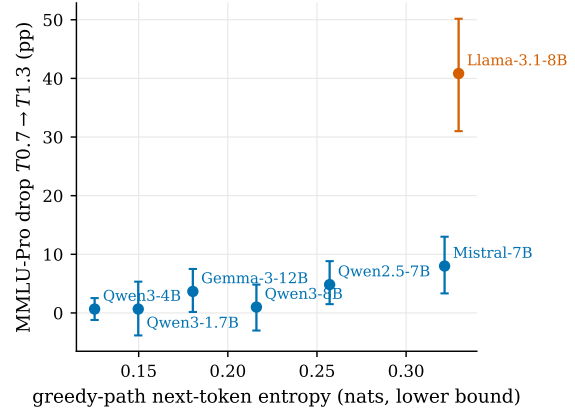


Figure 7: Per-token uncertainty on the model’s own greedy path versus the temperature drop (MMLU-Pro, 95% CIs). The rank correlation is $\rho = +0.96$; the Llama–Mistral gap at equal entropy shows flatness alone does not set the magnitude.

Sampler values: top- p 0.95, top- k 40, min- p 0.05, top- $n\sigma$ 1.0 (each method’s recommended value).

- **Seeding.** Every generation’s seed is derived deterministically from (global seed 0, model, quantization, sampler, temperature, task, item, repetition). Greedy decoding is the deterministic $T=0$ anchor with $K=1$.
- **Tasks.** GSM8K (openai/gsm8k, test split, first 50 items) and MMLU-Pro (TIGER-Lab/MMLU-Pro, test split, first 50 items; stratified replication in Appendix C). Parsers mirror lm-eval-harness; parse failures grade as incorrect and are reported.
- **Records.** Each generation is one JSONL record carrying the full identity tuple, the request parameters, the raw output, the parse path, and throughput covariates; runs are resumable by record id. Batched serving (≤ 8 slots) is seed-logged but not bit-exact.

- **Code and data.** [TODO: anonymized repository link for submission.]

SHA-256 of the exact rendered prompt.

- **Quantizations.** Every quantized checkpoint is a Bartowski imatrix GGUF (one provider, one recipe, so the quantization axis is not confounded by recipe). SHA-256 hashes of all 29 model files are pinned in the repository (DOWNLOADS.md and per-file .meta.json sidecars).

- **Decoding.** Exactly one truncation parameter per request; the sampler chain order is pinned per arm and recorded in every record.

Table 10: Accuracy (%) with temperature applied before (left) versus after (right) truncation, pooled over quantization.

Model	Method	temperature-first			temperature-last		
		0.7	1.0	1.3	0.7	1.0	1.3
<i>GSM8K</i>							
Llama-3.1-8B	top- <i>p</i>	81.5	78.0	63.8	80.8	78.7	73.0
Llama-3.1-8B	min- <i>p</i>	81.2	78.0	72.0	78.7	81.3	76.2
Mistral-7B	top- <i>p</i>	28.2	31.0	34.2	29.5	31.2	29.5
Mistral-7B	min- <i>p</i>	32.3	28.3	28.8	29.5	32.0	30.5
Qwen2.5-7B	top- <i>p</i>	92.2	91.2	90.3	90.8	91.3	90.3
Qwen2.5-7B	min- <i>p</i>	90.8	90.5	91.0	92.0	90.7	91.3
Gemma-3-12B	top- <i>p</i>	91.5	90.3	90.7	90.2	91.8	91.5
Gemma-3-12B	min- <i>p</i>	91.5	91.3	92.3	91.2	90.8	90.3
Qwen3-1.7B	top- <i>p</i>	73.8	75.0	72.7	73.0	72.0	71.3
Qwen3-1.7B	min- <i>p</i>	72.2	71.7	74.0	74.7	73.3	74.5
Qwen3-4B	top- <i>p</i>	95.5	95.5	93.5	94.8	95.2	94.3
Qwen3-4B	min- <i>p</i>	95.1	94.7	92.9	94.5	93.9	93.9
Qwen3-8B	top- <i>p</i>	94.0	94.3	93.7	93.8	94.0	94.0
Qwen3-8B	min- <i>p</i>	94.8	95.3	94.7	96.2	94.7	95.3
<i>MMLU-Pro</i>							
Llama-3.1-8B	top- <i>p</i>	51.3	46.0	16.0	50.3	46.7	37.5
Llama-3.1-8B	min- <i>p</i>	51.2	50.8	39.5	51.2	50.2	43.5
Mistral-7B	top- <i>p</i>	30.2	29.5	25.3	29.2	30.2	32.5
Mistral-7B	min- <i>p</i>	33.8	30.7	31.0	33.2	33.3	28.7
Qwen2.5-7B	top- <i>p</i>	65.5	62.8	62.7	63.7	62.7	61.3
Qwen2.5-7B	min- <i>p</i>	61.3	63.5	59.7	64.2	63.0	62.3
Gemma-3-12B	top- <i>p</i>	65.7	66.7	63.8	65.8	65.0	65.2
Gemma-3-12B	min- <i>p</i>	65.3	67.0	65.5	65.3	63.7	63.8
Qwen3-1.7B	top- <i>p</i>	47.2	47.0	47.2	47.7	46.8	43.5
Qwen3-1.7B	min- <i>p</i>	47.0	47.7	46.2	45.7	46.2	49.0
Qwen3-4B	top- <i>p</i>	62.8	61.3	62.8	62.9	62.7	61.7
Qwen3-4B	min- <i>p</i>	60.9	62.7	62.9	61.3	63.6	61.5
Qwen3-8B	top- <i>p</i>	60.5	61.3	60.5	60.7	59.5	60.5
Qwen3-8B	min- <i>p</i>	61.0	59.2	59.7	61.3	60.7	59.3

Table 12: Mean distinct parsed answers per item ($K=3$, pooled over quantization).

Model	GSM8K		MMLU-Pro	
	$T=0.7$	$T=1.3$	$T=0.7$	$T=1.3$
Llama-3.1-8B	1.51	2.33	1.71	1.28
Mistral-7B	2.21	2.48	1.97	2.21
Qwen2.5-7B	1.10	1.27	1.44	1.57
Gemma-3-12B	1.12	1.16	1.33	1.45
Qwen3-1.7B	1.41	1.43	1.67	1.73
Qwen3-4B	1.05	1.14	1.33	1.38
Qwen3-8B	1.09	1.13	1.41	1.40

Table 11: Plain-temperature accuracy drop $T_{0.7} \rightarrow 1.3$ (pp) per quantization level.

Model	Q8_0	Q6_K	Q4_K_M	Q3_K_M
<i>GSM8K</i>				
Llama-3.1-8B	+30.0	+32.0	+36.0	+38.0
Mistral-7B	-0.7	-0.7	+3.3	+12.0
Qwen2.5-7B	+4.0	+2.7	+2.7	+6.7
Gemma-3-12B	+3.3	+2.7	+0.0	+2.0
Qwen3-1.7B	+5.3	-6.0	-0.7	+0.0
Qwen3-4B	+2.7	+0.7	+6.0	+2.7
Qwen3-8B	-2.0	+1.3	+2.0	+0.7
<i>MMLU-Pro</i>				
Llama-3.1-8B	+38.0	+46.7	+38.0	+40.7
Mistral-7B	+9.3	+5.3	+10.7	+6.7
Qwen2.5-7B	+7.3	+6.0	+4.0	+2.0
Gemma-3-12B	+3.3	+5.3	+2.7	+3.3
Qwen3-1.7B	+0.7	+6.0	-8.0	+4.0
Qwen3-4B	-0.7	+0.0	+2.0	+4.0
Qwen3-8B	+2.7	-0.7	+0.0	+2.0

Table 13: Mechanism probes per model (Q8, 100 frozen prompts). Escape hazards are at $T=1.3$; drop is the MMLU-Pro plain-temperature $T_{0.7} \rightarrow 1.3$ accuracy drop (pp), pooled over quantization.

Model	H_{lb}	flat%	esc	Δ_{esc}	ΔH	drop
Llama-3.1-8B	0.329	7.1	3.26	+4.20	+0.110	+40.8
Mistral-7B	0.322	6.8	3.40	+1.19	+0.044	+8.0
Qwen2.5-7B	0.257	4.8	1.97	+1.00	+0.049	+4.8
Gemma-3-12B	0.180	3.0	1.44	+1.42	+0.055	+3.7
Qwen3-1.7B	0.150	2.1	0.36	-0.00	+0.012	+0.7
Qwen3-4B	0.125	1.5	0.22	+0.21	+0.021	+0.7
Qwen3-8B	0.216	3.8	0.71	+0.22	+0.024	+1.0